9.2 GPT4All

Probably the easiest way to try out local LLMs is the open-source program *GPT4All* from Nomic AI. The aim of the project is to run LLMs on local hardware (notebooks or desktop PCs) while offering a beginner-friendly interface. Another advantage is that the hardware requirements are relatively low and the application can run on a wide range of hardware. The most important features of GPT4All include the following:

- **Privacy**
  GPT4All attaches great importance to data privacy. The application models are executed locally on your system so that all input data, processing steps, and outputs don't leave the local machine. This minimizes the risk of sensitive data being intercepted by third parties.
- **Offline capability**
  GPT4All is an open-source project designed as a desktop application for Windows, macOS, and Ubuntu. After the installation, you can install LLMs locally and use them in the graphical user interface (GUI).
- **Many models available**
  GPT4All isn't tied to a specific LLM, but supports a variety of models such as Llama, Mistral, and Nous-Hermes. You can easily select and use the models in the application interface.
- **User-friendly interface**
  The GPT4All UI is designed to be simple and self-explanatory so that you can interact with it directly without further explanation.

9.2.1 Installation and Use

GPT4All is an open-source desktop application that you can install on your system to interact with locally installed LLMs. You can download installers for Windows, macOS, and Ubuntu from **www.nomic.ai/gpt4all**. After installation, you can launch the application and use it immediately.

The core feature is the chat, which works in a similar way to the commercial products ChatGPT, Gemini, and Claude. The big difference is that you must first download and integrate a model before entering your first prompt. You can add new models either directly on the start page via **Find Models** or via the navigation under **Models**. If you use the regular navigation, you'll first see a list of the locally installed models. Click the **Add Model** button to go to the model search and install a new model.

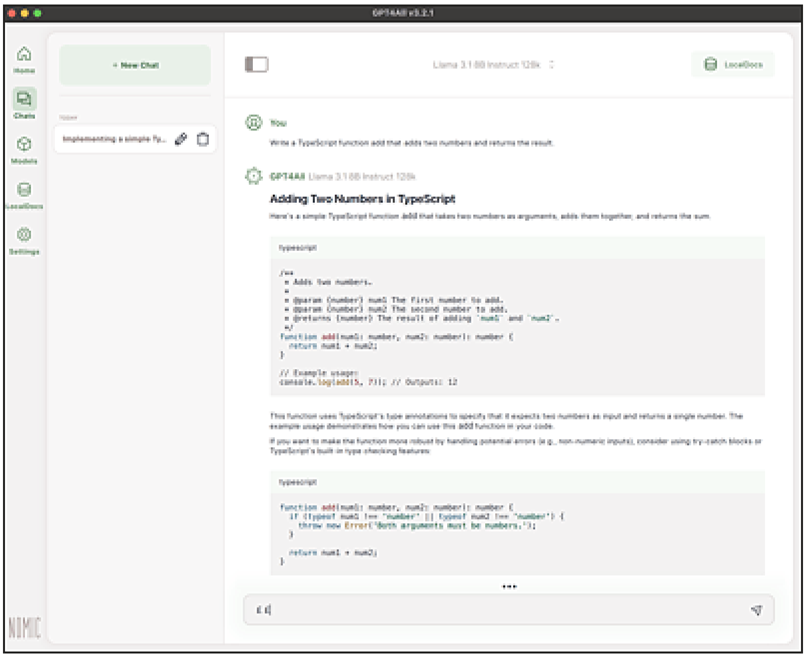

Figure 9.1 User Interface of GPT4All

GPT4All provides a brief description and the most important key figures for each model:

- **File size**
  When using local LLMs, you should have enough free space on your system's hard disk. The smaller models are in the single-digit gigabyte range, while the larger versions of Meta's Llama model already require more than 100 GB of memory. This storage space applies to the model and its weights. Larger models can depict complex relationships, but they also require more storage space.
- **RAM required**
  You need RAM to run a model. The more extensive the model, the more RAM you need to provide to avoid slowdowns or crashes.
- **Parameters**
  The parameters include the weights and biases of an LLM. They control how the model processes language and generates output. The more pa-

rameters an LLM has, the better it's able to process complex issues and generate better output. The disadvantage of a high number of parameters is that the model becomes significantly more computationally intensive and memory-intensive.

- **Quant**
  The quantization indicates the precision of the model weights. Internally, many models work with 32-bit floating point numbers. The accuracy can be reduced by quantization. Frequently used values are 4, 8, or 16 bits. Although this reduction is at the expense of precision, it increases speed and reduces memory requirements.
- **Type**
  The model type indicates the underlying model. Possible types include LLaMA3, Mistral, or GPT.

Once you've decided on one or more models, you can download them. Once the download is complete, GPT4All will provide a dropdown menu in the chat window where you can select the desired model and then use it for the current chat session.

9.2.2 Chat with Your Files

Another interesting feature of GPT4All is hidden behind the **LocalDocs** navigation item. GPT4All allows you to integrate your local documents and use them for queries. As the application runs completely locally, your documents don't leave the system and are therefore secure.

To integrate your documents, you should create a document collection by specifying a directory on your system in which the documents are located. GPT4All processes all documents in this directory and then makes them available for queries. For a new chat, you then select the model to be used and the document collection for this chat via the **LocalDocs** button. You can then formulate your prompts as usual, and GPT4All will take into account the documents in the collection. If you get a hit, you'll also receive a link to the relevant document.

The LocalDocs feature works via embeddings (see also Chapter 12). For this purpose, GPT4All stores the data from the documents in vectors and records their meaning. With the help of these vectors, the application is able to provide answers to questions about the documents very quickly. Internally, GPT4All uses the embedding models from Nomic AI, the manufacturer of GPT4All. The application then integrates the text passages found into the prompts to make the output more accurate and informative.

GPT4All supports plain text, PDF, and Markdown as document formats. When using GPT4All in development, this means that you can, for example, integrate books on design patterns or architecture as LocalDocs, or you can use the documentation of a programming language or framework directly to generate specific answers. This also allows you to avoid the disadvantage that LLMs usually don't know the latest development status of frameworks and libraries.

9.2.3 Conclusion

GPT4All is a convenient solution for getting started with local language models. It provides you with a user-friendly interface and allows you to add new models. In addition to the models you find via the search, you can also install and use your own models. A very useful feature is LocalDocs, which allows you to easily integrate local documents into your prompts without having to worry about the underlying mechanisms.

In addition to these features, GPT4All also offers you the option of accessing the application via various language bindings, so that you can access a local language model in a Python or JavaScript application, for example.

GPT4All is intended for getting started with local language models. Sooner or later, you'll reach the limits of the application, especially in a professional environment. In the rest of this chapter, you'll learn about alternatives to GPT4All for local language models, which are somewhat more complex to set up and configure but give you better control.